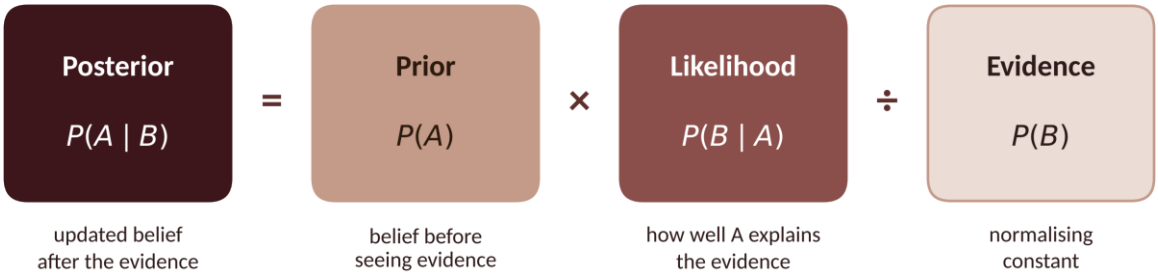
PART A · THEORY

Probability & Conceptual Foundation

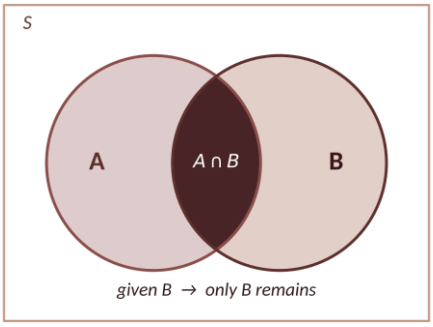


1 What is Conditional Probability?

Conditional probability is the probability that event A occurs **given that** event B has already occurred. Knowing B shrinks the sample space from S to B.

$$P(A \mid B) = P(A \cap B) / P(B), \quad P(B) > 0$$

Example: P(spam | message contains “free”).



2 Explain Bayes’ Theorem and its importance in classification problems.

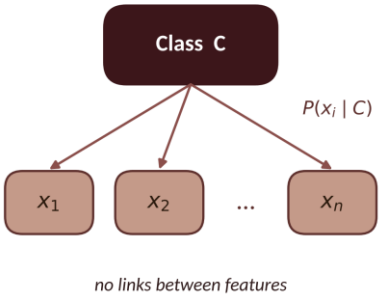
Bayes’ theorem updates the probability of a hypothesis A after observing evidence B:

$$P(A \mid B) = P(B \mid A) \cdot P(A) / P(B)$$

Importance: it converts “how likely is this data, given the class?” into “how likely is the class, given this data?”. A classifier computes this posterior for every class (A = class, B = observed features) and predicts the highest one. It is the foundation of Naive Bayes and of probabilistic models such as spam filters.

3 What assumptions does the Naive Bayes Classifier make?

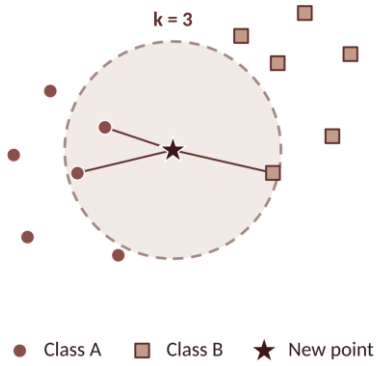
- **Conditional independence:** given the class, features do not depend on each other, so P(X | C) = P(x1 | C) × P(x2 | C) × … × P(xn | C).
- **Equal importance:** every feature contributes independently to the prediction.
- **Assumed distribution:** features follow a chosen form (Gaussian, Multinomial or Bernoulli) and class priors are estimated from training data.



4 Explain the working principle of KNN and SVM.

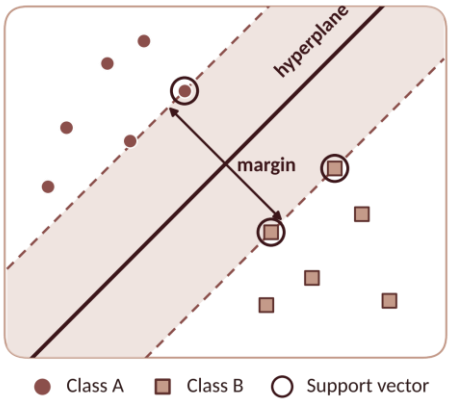
K-Nearest Neighbors (KNN)

A **lazy, distance-based** learner. For a new point it measures the distance (usually Euclidean) to all training points, picks the **k nearest neighbours** and assigns their **majority class**. Features must be scaled; k balances noise against over-smoothing.



Support Vector Machine (SVM)

Finds the **hyperplane with the maximum margin** between classes; only the nearest points, the **support vectors**, define it. A soft margin (C) tolerates errors, and kernels (RBF, polynomial) handle non-linear data.



5 Compare distance-based, probabilistic, and margin-based classifiers.

	Distance-based KNN	**Probabilistic** Naive Bayes	**Margin-based** SVM
Core idea	Similar points share a class	Pick the class with the highest posterior probability	Separate classes with the widest possible margin
Training	None (lazy learner)	Very fast: counts and estimates	Heavier: solves an optimisation problem
Output	Majority vote of k neighbours	Class probabilities	Side of the boundary (score)
Strength	Simple; flexible, non-linear boundaries	Fast; works with little data and text	Strong in high dimensions; robust margin
Limitation	Slow prediction; needs scaled features	Feature independence is rarely true	Slow on huge data; kernel and C need tuning

In short: KNN suits small, well-scaled data; Naive Bayes is the fastest and works well for text such as spam; SVM is strongest when classes are separated by a clear margin.